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The Accuracy of Toyota Vehicle Control History Data during Autonomous Emergency Braking

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Abstract

Newer Toyota vehicles store information about more than 50 parameters for 5 s before and after non-collision events in the Vehicle Control History (VCH) records. The goals of this study were to assess the accuracy of VCH data acquired during Autonomous Emergency Braking (AEB) events and to investigate the effects of speed, acceleration, and system settings on AEB performance. A 2017 Toyota Corolla with Safety Sense P Pre-Collision System (PCS) was driven in a straight line towards a car-like target at different combinations of four speeds (20, 25, 30, and 40 km/h; or 12, 15, 19, and 25 mph) and three accelerator pedal positions (constant 30%, 40%, and 50% accelerator opening ratios) until the AEB system activated. The vehicle speed, vehicle acceleration, radar target closing speed, and radar target distance recorded in the VCH were compared to a reference 5th wheel. We found that errors in the VCH distance, speed, and acceleration data varied with the test conditions. Regression equations were derived to better predict distance, speed, and acceleration from the VCH data. A driver-adjustable PCS warning setting only altered the timing of the warning and not the underlying AEB response. The vehicle struck the target most often at 20 km/h (12 mph) when accelerating towards the target, but did not strike the target when approaching it at a constant speed of 20 km/h. This study serves as an initial investigation into the accuracy of VCH data and the performance of the Toyota PCS under various conditions and settings.

Introduction

Accident investigators routinely rely on crash data read from airbag control modules using the Bosch CDR System [1, 2, 3, 4, 5]. Many newer Toyota vehicles also store information about non-crash events in the Vehicle Control History (VCH) records [6]. VCH records can be triggered by over 20 types of events, including activation of the Antilock Braking System (ABS) and Pre-Collision System (PCS) [6].

The Toyota PCS uses a millimeter wave radar and a monocular camera to detect potential hazards and initiate Autonomous Emergency Braking (AEB) [7]. In the 2017 Toyota Corolla, the radar unit is mounted behind the front grille and the monocular camera is mounted inside the front windshield ahead of the rear-view mirror. When the PCS detects an impending collision, an audible alarm sounds, and a "BRAKE" message is displayed on the instrument panel. If required, the PCS can apply the brakes autonomously and bring the vehicle to a stop. In the Toyota Corolla, there are 4 stages to the PCS response: alarm (ALM), pre-collision brake assist (PBA), former pre-collision brake (FPB), and pre-collision brake (PB). The first stage (ALM) is the simultaneous activation of an audible alarm and a "BRAKE" message on the instrument panel. The second stage (PBA) is the activation of a pre-collision brake assist, which amplifies the driver's braking level by increasing amounts as the time-to-impact (TTI) decreases. The third and fourth stages (FPB and PB) are autonomous braking by the PCS. Using steering wheel buttons to access the settings menu in the multi-information display, the "pre-collision warning timing" setting can be changed between three available settings. An icon in the shape of three concentric semi-circular bars is used to select the settings, with three shaded bars representing the "far" setting, two shaded bars representing the "middle" setting (described as the default setting in the Corolla's owner's manual), and one shaded bar representing the "near" setting.

The VCH record can contain over 50 vehicle parameters for 5 s before and after a triggering event (Appendix A) [6]. Vehicle speed, engine RPM, accelerator opening ratio (a measure of accelerator pedal position), and brake switch status are recorded for all triggering events. Specific events may contain additional data. For instance, ABS events include wheel speed sensor data and PCS events include target radar distance and target radar relative speed data.

VCH data include some of the same pre-crash data found in collision records downloaded using the Bosch CDR System. Ruth et al [8] tested the accuracy of the vehicle speed data in the CDR records of 2010 and 2011 Toyota Camrys and found the accuracy varied from -3.0 to +0.4 km/h under steady speed conditions and -7 to +15 km/h under heavy braking. Ruth et al found the CDR reported speed was rounded down to the next lower even integer value (in km/h). The pre-crash vehicle speeds were retrieved over the vehicle's Controller Area

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Network (CAN) bus. Ruth et al reported that speed information on the vehicle CAN bus and the speed data in the collision records are updated asynchronously at 2 Hz. Therefore, the CDR data can underestimate the vehicle speed during acceleration and overestimate it during deceleration. Wheel slip during acceleration and deceleration counteracted some of this effect. Brown and White [9] found similar biases during acceleration and deceleration when comparing CAN bus data of a 2010 Toyota Camry to external GPS and accelerometers. Brown and White found the vehicle CAN bus data underreported speed by 2.7% during steady state speeds.

The goals of this study were to i) determine the accuracy of the radar distance, radar velocity, vehicle speed sensor, and longitudinal accelerometer data in VCH records triggered by PCS and ABS events, ii) quantify the effect of initial distance, speed, and acceleration on PCS performance, and iii) quantify the effect of the "pre-collision warning timing" setting on PCS performance. Our test results will help accident investigators analyze incidents involving vehicles equipped with AEB systems.

Methodology

Test Vehicle

A 2017 Toyota Corolla (VIN: 2T1BURHE1HC797605) equipped with an automatic transmission and the Toyota Safety Sense P package was used for all tests (Figure 1). There was no damage or dirt on the radar unit, front grille, or windshield ahead of the vehicle's camera. The vehicle was fitted with manufacturer specified 205/55R16 tires that were inflated to the manufacturer recommended 32 psi. A video camera (GoPro Hero 3+, San Mateo, CA) was mounted near the center of the ceiling and aimed forward to capture the driver's actions and a view through the windshield.

A custom-made bracket was put under the accelerator pedal assembly to limit its movement and achieve a constant position for each test. Accelerator opening ratio, a parameter in the VCH records, was used as an indicator of accelerator pedal position. Brackets were built to limit the accelerator pedal movement to achieve 30%, 40%, and 50% accelerator opening ratios, which were verified using the Toyota Techstream tool through the Data List function. The "pre-collision warning timing" setting on the multi-information display was set to "middle" of the three available settings ("far", "middle", "near") for all tests except those designed to evaluate the effect of this setting.

The VCH records in the test vehicle included PCS events that were triggered by the onset of the PCS ALM flag. Data within the PCS events are recorded at 2 Hz and are supplemented with asynchronous data when PBA, FPB, and PB flags occur. The resolution of the times associated with these flags was 0.01 s. The Corolla VCH can store up to 4 PCS events before overwriting the earliest event. For ABS events, the VCH is triggered by the onset of the ABS and data are recorded at 6.67 Hz. The Corolla VCH can store up to 5 ABS events before overwriting the earliest event. The vehicle's PCS alarm buzzer voltage was monitored to record the onset of the PCS alarm and used to synchronize the 5th wheel data to the VCH data. The left rear tail light voltage was also monitored to record the onset of brake lamp illumination.

FIGURE 1 2017 Toyota Corolla with Toyota Safety Sense P.



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Autonomous Emergency Braking (AEB) Target

The AEB target consisted of a 6-inch thick open-cell foam slab (1.8 lbs/ft^3, The Foam Shop, Richmond, BC) cut into the cross-sectional shape of a 2010 Hyundai Sonata and covered with vinyl on which was printed an image of the Sonata (2905 x 2240 resolution). Retro-reflective tape (Reflective Conspicuity Tape, 3M, St. Paul, MN) was adhered to the tail lights and brake lights to mimic their reflective behavior, and two 15 × 136 cm strips of 100% polyvinyl chloride lamented polyester fabric (Energy Shield 200 - 10 Oz, Online Fabric Store, West Springfield, MA) were glued to the foam under the vinyl cover at heights of 58 and 88 cm (measured from the bottom, Figure 2). The overall

FIGURE 2 Photograph of the AEB target showing the locations of the radar reflective strips (black dashed rectangles) and the corner radar reflector (white dashed triangle).



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dimensions were 183 cm wide by 152 cm tall, consistent with the Euro NCAP target dimensions [10].

The target was supported by a lightweight frame constructed from 1-inch IPS Series 200 PVC piping. The target was configured with a trihedral triangular corner radar reflector mounted at the target centerline at a height of 64 cm (the height of the Corolla's radar unit, Figure 2). The radar reflector had inside edge lengths of 55 mm, which at 76.5 GHz represented a radar cross-section of about 2.5 m^2, consistent with the Euro NCAP's radar reflector [10].

Instrumentation

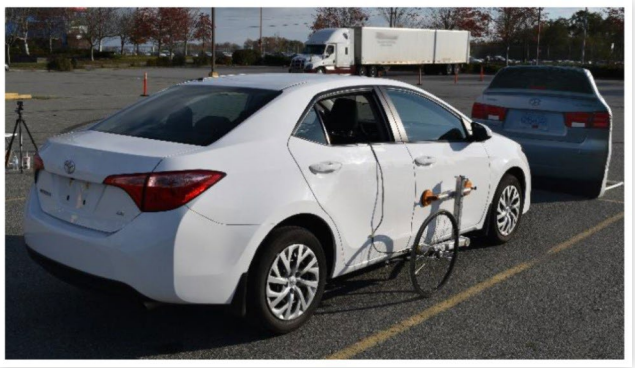
A calibrated 5th wheel (MEA Forensic, Richmond, BC) was mounted to the right side of the test vehicle to measure its travel distance and to compute its velocity and acceleration (Figure 3). The 5th wheel's distance accuracy was previously reported to be within 0.1% [3] and its speed resolution was previously reported to be within 0.01 m/s [11]. The 5th wheel was sampled at 256 Hz and data were collected for 16 seconds post trigger. A DL1 datalogger (Race Technology Ltd, Nottingham, UK) recorded the 5th wheel trigger signal, PCS alarm buzzer voltage, and brake lamp voltage. The DL1 was manually triggered first, followed by the 5th wheel. The DL1 data were sampled at 1000 Hz and data were collected until the vehicle came to a stop.

Test Procedure

All tests were conducted on a large flat asphalt parking lot over two dry and sunny days. The centerlines of the test vehicle and AEB target were aligned and the test vehicle was driven in a straight line towards the target (Figure 3). The target was oriented so that the surface facing the test vehicle remained shaded. The asphalt surface extended 50 m past the target location. The target background consisted of low lying shrubs, large deciduous trees, power poles, light standards, and a chain-link fence (Figure 4).

One person (author Xing) drove the vehicle for all tests. The accelerator pedal was pushed to, and held against the limiter bracket until the vehicle stopped or hit the target. Steering was kept to a minimum. Following each test, the distance between the front bumper of the test vehicle and the front face of the target was measured (S_{Rest}). If the target was struck, the distance between the front bumper and the initial position of the target was measured and recorded as a negative number. VCH records related to PCS and ABS events were read every three tests through the vehicle's data link connector (DLC) port with a Mongoose Pro MFC2 cable (Drew Technologies, Ann Arbor, MI) and a laptop running Toyota Techstream software (version 12.20.024).

FIGURE 3 Test setup and 5th wheel.



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FIGURE 4 Test site background with the test vehicle and the AEB target in the foreground.



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A total of 104 tests were conducted in three sets. The first set, referred to as the "matrix tests", were used to assess the first two goals of this study. The matrix tests (n=82) were conducted using a combination of four test speeds (20, 25, 30, and 40 km/h) and three accelerator pedal limiters (30%, 40%, and 50% accelerator opening ratios). Start locations for each test speed and accelerator pedal position combination were set so that the vehicle reached the test speed when the PCS alarm sounded (Table 1). The same test speed and accelerator pedal limiter were used for a block of three consecutive tests, but the presentation order of these blocks was randomized. Our test speeds included 20 km/h (12 mph) and 40 km/h (25 mph), which are used by the Insurance Institute for Highway Safety (IIHS) in their AEB ratings tests [12].

The second set of tests, referred to as the "radar warning tests", were used to assess the third goal of this study. The radar warning tests (n=22, 16 new tests combined with 6 tests from the matrix tests) were conducted using three different "pre-collision radar warning timing" settings ("far", "middle", and "near"). These tests were conducted at a test speed of 40 km/h using a 30% accelerator position. The presentation order of the warning settings was randomized in blocks of three similar consecutive tests.

TABLE 1 Starting distances for matrix tests.

Accelerator Opening Ratio	Test Speed (km/h)			
	20	25	30	40
30%	20.45 m	28.90 m	40.80 m	76.20 m
40%	19.55 m	27.10 m	36.70 m	67.20 m
50%	18.55 m	25.60 m	33.25 m	54.80 m

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The third set of tests, referred to as the "long approach tests", were used to explore whether the methods used in the matrix tests were responsible for the high rate of target strikes we observed at the 20 km/h test speed. The long approach tests (n=6) were conducted at a test speed of 20 km/h, using a 30% accelerator position, and a start position (76 meters from the target) that allowed a constant speed to be held for about 6 seconds prior to the PCS alarm. The accelerator pedal was pushed to and held against the limiter bracket until the vehicle reached 20 km/h, then the accelerator pedal was modulated by the driver to maintain 20 km/h.

For all three experiments except for the 20 km/h matrix tests, tests were conducted until a minimum of 6 tests with no target trikes were achieved for each test condition. After each block of 3 tests, the vehicle ignition was cycled to increment the ignition counter, which served as a test block identifier.

Data Extraction and Reduction

The following information was extracted from each VCH PCS event record, which was recorded at 0.5 s increments, and at each asynchronous PCS flag trigger related to the ALM, PBA, FPB, and PB flags:

- Time (s) - time since ignition on
- Vehicle Speed (V_{VSS}) (km/h) - Vehicle speed sensor speed
- Longitudinal G Sensor Signal (A_{SENSOR}) (m/s^2) - longitudinal acceleration
- ALM Request Flag (0/1) - PCS request flag for alarm
- PBA Request Flag (0/1) - PCS request flag for pre-collision brake assist
- FPB Request Flag (0/1) - PCS request flag for former pre-collision brake
- PB Request Flag (0/1) - PCS request flag for pre-collision brake
- Target Distance (S_{RADAR}) (m) - radar-based distance to target
- Target Relative Velocity (V_{RADAR}) (m/s) - radar-based target closing speed

For each matrix test, we also looked for a corresponding ABS event in the VCH. If an ABS event was present, we extracted the following data, which were recorded at 0.15 s increments:

- Time (s) - time since ignition on
- Under ABS Control (0/1) - ABS intervention status
- Vehicle Speed (V_{VSS}) (km/h) - Vehicle speed sensor speed
- Longitudinal G Sensor Signal (A_{SENSOR}) (m/s^2) - longitudinal acceleration

We calculated TTI values for the PCS flags (TTI_{ALM}, TTI_{PBA}, $TTI_{FPB,}$ and TTI_{PB}) by dividing the Target Distance (S_{RADAR}) by the Target Relative Velocity (V_{RADAR}) at each corresponding time sample in the VCH data (Equation 1).

$$TTI = \frac{Target\ Distance\left(S_{RADAR}\right)}{Target\ Relative\ Velocity\left(V_{RADAR}\right)} \quad (1)$$

The 5th wheel produced distance, velocity, and acceleration data for each test. The distance travelled was converted to distance from the target (S_{MEA5W}) by adding the total distance travelled and the distance to target at rest, and then subtracting distance travelled for each time sample. The velocity data (V_{MEA5W}) were calculated using finite differences across 3 distance samples and then filtered using a 4th-order 5 Hz Butterworth low-pass filter. The acceleration data (A_{MEA5W}) were calculated using an 81-point (316 ms) moving average, which optimally reduced the effect of vehicle pitch oscillations on the calculated acceleration level.

The VCH data were synchronized with the DL1 data using the start of the PCS alarm, which appeared in both data sets. Time zero was defined as when the ALM flag appeared in the VCH PCS data and when the PCS alarm buzzer sounded, indicated by a sudden voltage change in the DL1 data. The 5th wheel data were synchronized with the VCH data using the time difference between the 5th wheel trigger and the start of the PCS alarm as recorded by the DL1.

Data from the matrix tests were used to assess the accuracy of the VCH radar distance (S_{RADAR}), radar speed (V_{RADAR}), and longitudinal G sensor (A_{SENSOR}) data. We time-matched the reference 5th wheel data to all samples within the VCH record (typically 24 data points per record). Since the Toyota Corolla's owner's manual indicated that the PCS system is active between 10 and 180 km/h, samples where the radar speed (V_{RADAR}) was below 10 km/h were removed from the analysis. There were four instances when the vehicle speed was above 10 km/h and where the difference between the radar distance and 5th wheel distance exceeded 15 meters, which we interpreted as the PCS prioritizing a different target. These four data points (three from one test, one from a second test) were removed from the analysis. The remaining differences between the radar distance and 5th wheel distance were clustered between 0 and 4 m. If the vehicle struck the AEB target, then all VCH samples after the target strike were removed. The VCH reported radar target relative velocity as a negative number, which we converted to a positive number and referred to as radar speed (V_{RADAR}).

Vehicle Speed Sensor (VSS) speed (V_{VSS}) data from PCS and ABS events were time-matched to the 5th wheel data. For each VSS speed in the ABS events, we also recorded whether the ABS was active ("Under ABS Control" flag was 1).

We compared the distance (S_{RADAR}), speed (V_{RADAR} and V_{VSS}), and acceleration (A_{SENSOR}) values in the VCH records to the distance (S_{MEA5W}), speed (V_{MEA5W}), and acceleration (A_{MEA5W}) measured and calculated from the 5th wheel. We assumed the 5th wheel to be the reference system. Since our target was stationary, the relative radar speed (V_{RADAR}) was assumed to be an independent measure of absolute vehicle speed.

For the matrix tests, the VCH data (S_{RADAR}, V_{RADAR}, and A_{SENSOR}) and the difference between the VCH and 5th wheel data were first plotted against the 5th wheel distance, speed, and acceleration (Figure 6). Linear regression analyses were then performed to determine the best predictions of the actual distance, speed, and acceleration (as represented by the 5th

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wheel data) from the available VCH data (S_{RADAR}, V_{RADAR}, and A_{SENSOR}) (Equations 2 to 4). The $S_{predicted}$, $V_{predicted}$, and $A_{predicted}$ on the left side of Equations 2 to 4 represent the corrected variable of interest. On the right side of the equation, S_{RADAR}, V_{RADAR}, and A_{SENSOR} are variables retrieved from the VCH and β coefficients were determined by the linear regression analyses. Each β coefficient was tested to determine if it was significantly different from zero. Data points in which the VCH accelerations saturated (A_{SENSOR} = -8.9725 m/s^2) were removed prior to the regression analyses. For full regressions containing variable(s) with small β coefficients, reduced linear regression analyses were performed with the negligible variable(s) dropped and new β coefficients were determined for the reduced regression model.

Differences in the PCS flag TTIs were assessed using a two-way ANOVA (4 speed levels and 3 accelerator levels) and differences in the rate of target strikes were assessed using an omnibus χ^2 test followed by pairwise comparisons. For the radar warning tests, a separate one-way ANOVA with three levels (far, middle, near) was used to evaluate differences in the vehicle speed (V_{ALM}) at PCS alarm, target distance (S_{ALM}) at PCS alarm, each of the four PCS flag TTIs, and the target distance at rest (S_{Rest}). All statistical tests were performed in Statistica (v6.0, Statsoft, Tulsa, OK) using a significance level of $p<0.05$.

$$S_{predicted} = \beta_0 + \beta_1 S_{RADAR} + \beta_2 V_{RADAR} + \beta_3 A_{SENSOR} \quad (2)$$

$$V_{predicted} = \beta_0 + \beta_1 S_{RADAR} + \beta_2 V_{RADAR} + \beta_3 A_{SENSOR} \quad (3)$$

$$A_{predicted} = \beta_0 + \beta_1 S_{RADAR} + \beta_2 V_{RADAR} + \beta_3 A_{SENSOR} \quad (4)$$

Results

Target Strikes

There were 20 target strikes within the 82 matrix tests (Table 2). There were significantly more target strikes at the 20 km/h test speed than at the other three tests speeds ($p<0.0003$). There were no significant differences between the rate of target strikes at the three accelerator positions ($p>0.22$). There was one target strike within the 22 radar warning setting tests and no target strikes within the 6 long approach tests.

Matrix Tests

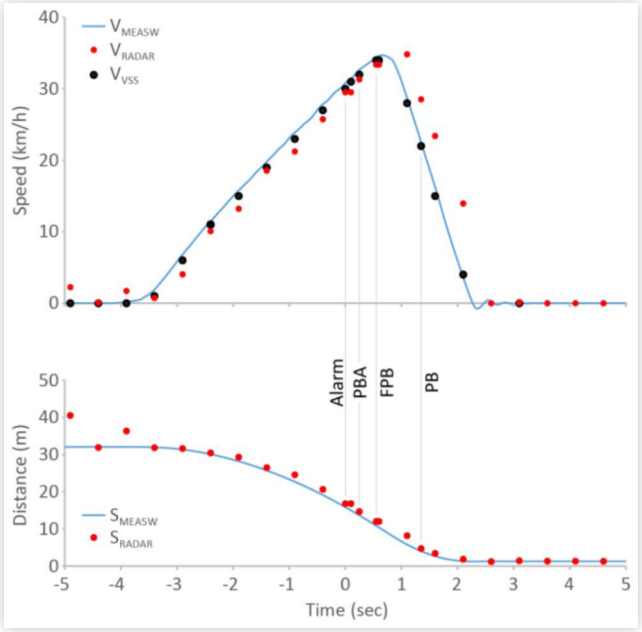
The radar-measured target distance (S_{RADAR}) and target relative velocity (V_{RADAR}) exhibited a consistent pattern of misalignment with the 5th wheel distance (S_{MEA5W}) and speed (V_{MEA5W}) data (Figure 5). The radar distance and speed lagged the 5th wheel and vehicle speed sensor (V_{VSS}) data when the vehicle accelerated or decelerated, with the lag becoming more apparent at the higher deceleration (Figure 5). The radar also periodically lost track of the AEB target, particular when further away from the AEB target. This occurred both when the vehicle was stationary and when it was moving. We observed distance errors of up to 70 m in some instances (although all those over 15 m were removed from the data analysis, leaving only distance errors ≤ 4 m in the analyzed data).

TABLE 2 Target strikes (numerator) versus the total number of tests (denominator) during the matrix tests.

Accelerator Opening Ratio	Test Speed (km/h)			
	20	25	30	40
30%	3/6	1/7	0/6	0/6
40%	5/6	0/6	2/9	0/6
50%	5/6	2/9	2/9	0/6

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FIGURE 5 Sample speed and distance data from one of the matrix tests showing the relationship between the 5th wheel data, the radar data, and the VSS data.



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The radar-measured target distance (S_{RADAR}) consistently overestimated the 5th wheel-measured distance to target (S_{MEA5W}) by 1.18 ± 0.60 m (max error 3.87 m; Figure 6, left column). The radar distance error (S_{RADAR} - S_{MEA5W}) increased with increasing closing velocity, but did not appear to vary markedly with acceleration (Figure 6). The regression model for predicting actual distance (defined as the 5th wheel distance) from the VCH data contained significant terms for radar distance, radar speed, and longitudinal G sensor acceleration (Table 3).

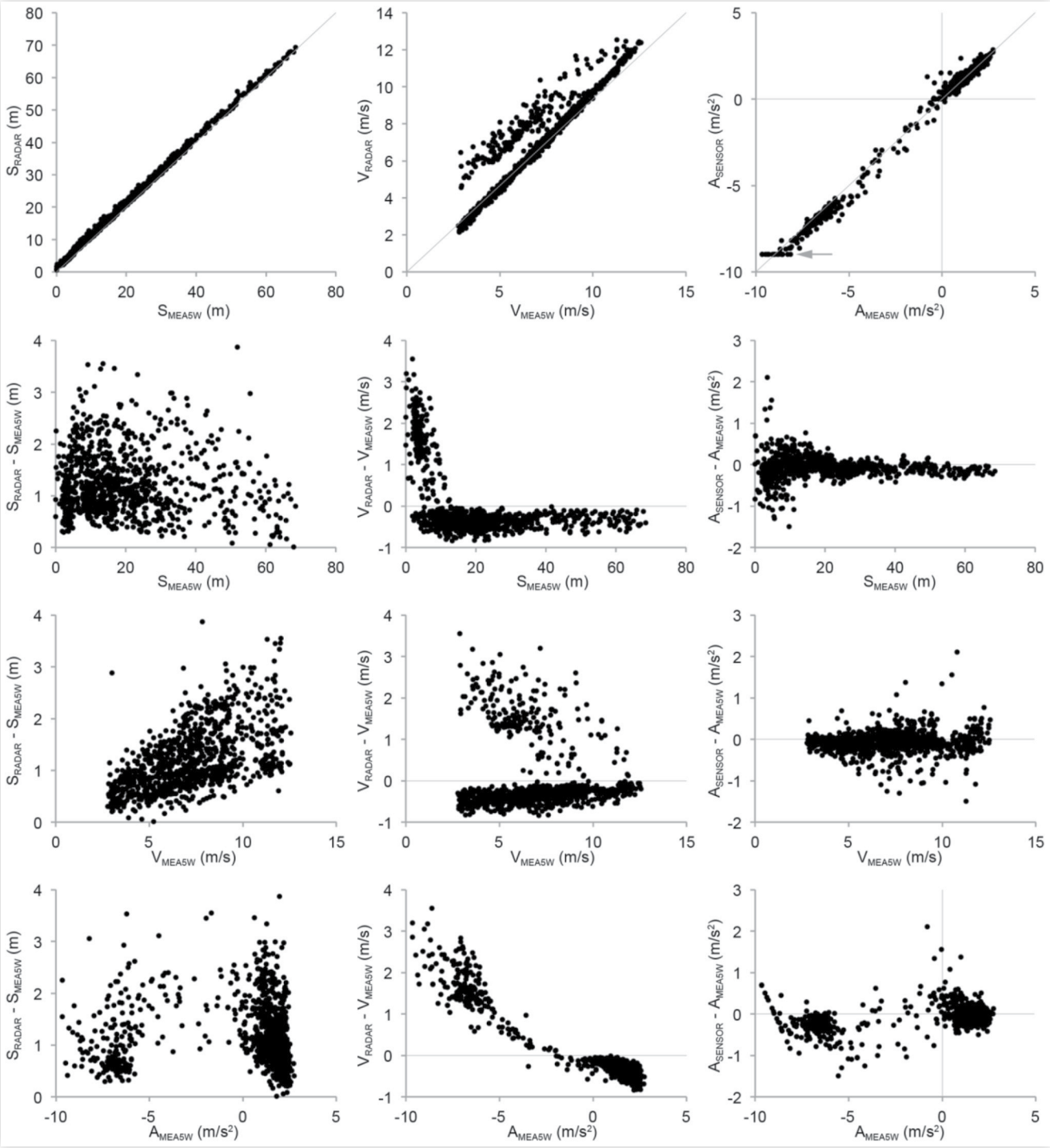
The radar-measured closing velocity (V_{RADAR}) consistently underestimated the actual speed while the vehicle accelerated (-0.36 ± 0.16 m/s) and consistently overestimated the speed while the vehicle braked (1.32 ± 0.84 m/s) (Figure 6, center column). The regression analysis yielded a model containing all three terms, however the intercept was not significantly different from zero (Table 3). A reduced model (excluding S_{RADAR}) yielded similar coefficients and a similar prediction interval for the estimated actual velocity.

The VCH longitudinal G sensor (A_{SENSOR}) saturated during braking in 16 of the 1047 data samples extracted from the 82 matrix tests (Figure 6, right column). Excluding the saturated

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FIGURE 6 Summary of matrix test results. The top row of graphs show comparisons of the radar distance, radar speed, and longitudinal G sensor acceleration (all from the VCH) versus the corresponding 5th wheel distance, speed, and acceleration. The diagonal line represents a 1:1 relationship and the grey arrow in the acceleration graph show the saturation point of the accelerometer recorded in the VCH. The lower nine graphs show how the difference between the VCH data and the 5th wheel data varies with the distance, speed, and acceleration measured by the 5th wheel.



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data, the longitudinal G sensor values were 0.063 ± 0.264 m/s^2 lower than the 5th wheel acceleration values. This resulted in higher magnitude deceleration values for the VCH data compared to the 5th wheel data. The regression analysis yielded a model that contained all three terms with a non-significant intercept (Table 3). A reduced model (excluding S_{RADAR} and V_{RADAR}) yielded a significant intercept and a slightly larger prediction interval on the estimated actual acceleration.

The VSS speed from PCS events (V_{VSS}) consistently underestimated the reference 5th wheel speed (V_{MEA5W}) (-0.12

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TABLE 3 Regression model coefficients, standard errors (in parentheses), p-values, and prediction interval (PI) of the estimates for the full models and reduced models (marked with a *).

Model	β_0	β_1	β_2	β_3	PI
Distance (m)	-0.1528	0.9980	-0.1364	-0.0183	±1.00
($S_{predicted}$)	(0.0587)	(0.0012)	(0.0068)	(0.0056)	
p-value	0.0094	<0.0001	<0.0001	0.0012	
Speed (m/s)	-0.0280	-0.0011	1.0082	0.2274	±0.46
($V_{predicted}$)	(0.0268)	(0.0006)	(0.0031)	(0.0026)	
p-value	0.2960	0.0495	<0.0001	<0.0001	
Speed*	-0.0525		1.0085	0.2251	±0.46
($V_{predicted, reduced}$)	(0.0237)		(0.0031)	(0.0023)	
p-value	0.0270		<0.0001	<0.0001	
Acceleration (m/s^2)	0.0065	0.0056	-0.0076	0.9489	±0.42
($A_{predicted}$)	(0.0254)	(0.0005)	(0.0029)	(0.0024)	
p-value	0.7969	<0.0001	0.0098	<0.0001	
Acceleration*	0.0637			0.9623	±0.47
($A_{predicted, reduced}$)	(0.0073)			(0.0022)	
p-value	<0.0001			<0.0001	

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± 0.16 m/s) (Figure 7). There were quantization errors in the VSS data because they were recorded to the next lowest km/h. The ABS activated in some of the tests, and this generated an ABS event record in addition to a PCS event record in the VCH. Matrix tests with ABS event data were extracted to quantify the effect of ABS intervention. The VSS data from the ABS events showed similar quantization errors, but underestimated the reference 5th wheel speed less during non-ABS intervention (-0.08 ± 0.16 m/s) than during ABS intervention (-0.56 ± 0.69 m/s) (Figure 7). The VSS underestimated the 5th wheel speed by a maximum of 8.1 km/h and 11.2 km/h for PCS and ABS event records respectively (Figure 7).

For 62 out of the 82 matrix tests, the test vehicle came to a stop before the target (Figure 8, Table 4). The distance at rest from the target (S_{Rest}) for these tests were 134.3 ± 15.6 cm. Accelerator

opening ratio had a significant effect on the distance at rest from the target (p=0.03), which was slightly longer for the 50% opening ratio (140.9 ± 24.5 cm) than for the 40% opening ratio (129.3 ± 4.4 cm). There was no effect of test speed on the distance at rest from the target. All four PCS flag TTIs varied with test speed, but only TTI_{ALM} and TTI_{FPB} varied with accelerator position. Despite the statistical difference in TTI_{PB} with test speed, there was considerably less variability in TTI_{PB} than in the other three TTIs (Figure 8).

The last recorded radar-measured distance-to-target value was compared to the hand-measured distance at rest from the target. The radar value consistently overestimated the measured distance to target by 5.28 ± 1.53 cm.

Radar Warning Tests

Changing the "pre-collision warning timing" between "far", "middle", and "near" resulted in significantly different speeds at PCS alarm (V_{ALM}), distance at PCS alarm (S_{ALM}), and PCS ALM flag TTI (Table 5). The speed differences were small and only present between the "middle" and "near" settings, and did not explain the larger differences between all three settings for the distance and TTI findings. There were no significant differences in the latter three TTIs (TTI_{PBA}, TTI_{FPB}, TTI_{PB}) or the distance to the target at rest (S_{Rest}).

Long Approach Tests

Six tests were conducted at a test speed of 20 km/h, using a 30% accelerator opening ratio, and a start position that allowed a constant speed to be held for about 6 seconds prior to the PCS alarm. All 6 tests had ALM, FPB, and PB flags, but none of the tests had a PBA flag. None of these 6 tests resulted in a target strike.

For the 18 matrix tests performed at 20 km/h, 13 tests resulted in a target strike and had ALM and PBA flags, but no FPB or PB flags. Four tests that did not result in a target strike had no PBA flags, but had FPB and PB flags (Figure 8). In another test that did not result in a target strike, the PBA flag occurred after the FPB flag (Figure 8).

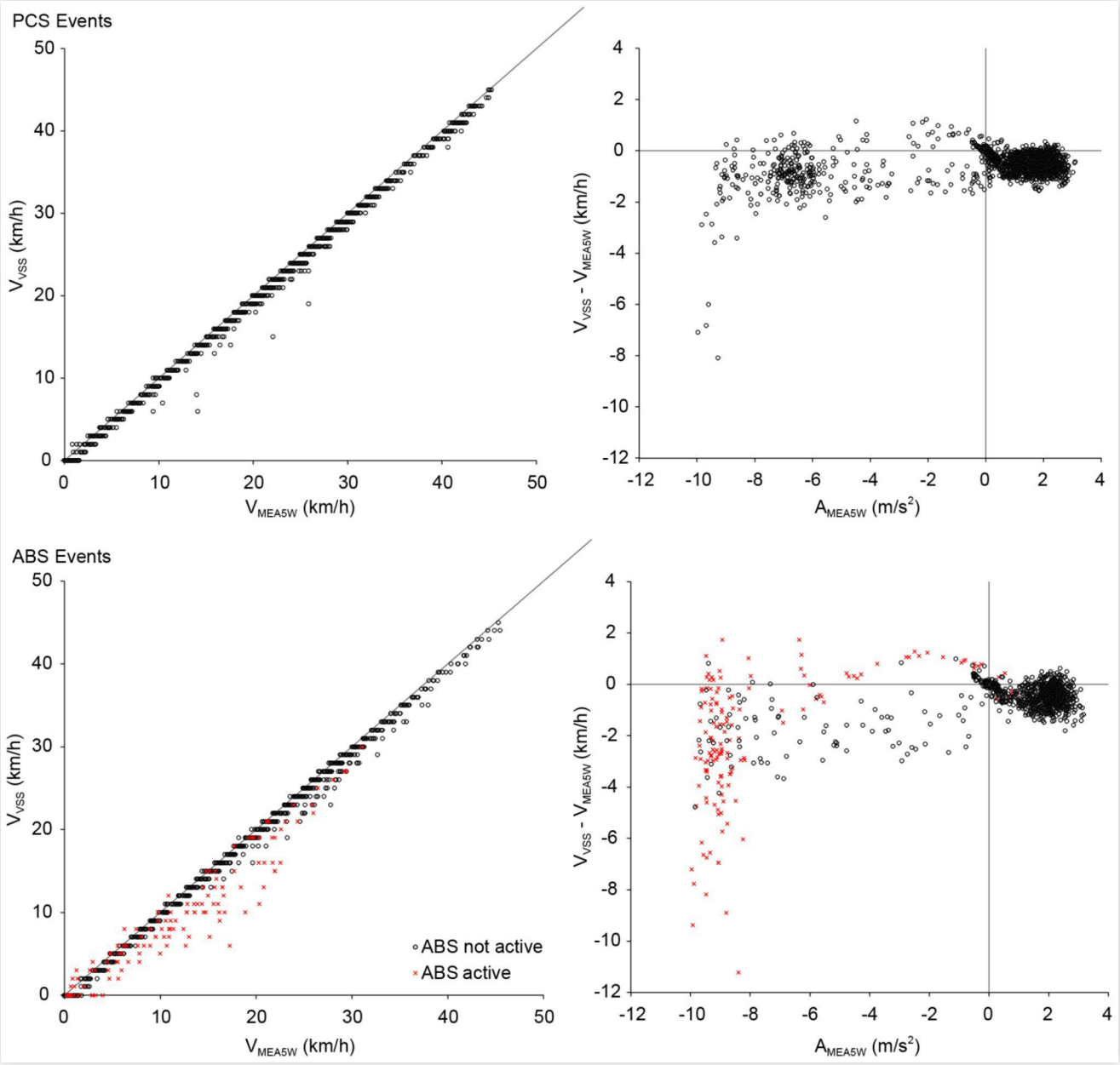
Discussion

Our three goals were to determine the accuracy of the VCH data in PCS events, to quantify the effect of initial vehicle kinematics on PCS performance, and to quantify the effect of the "pre-collision warning timing" setting on PCS performance. With respect to the accuracy of the VCH data, we found that radar distance, radar speed, and longitudinal G sensor acceleration data in VCH records were significantly different from reference 5th wheel data. Using the linear regression models presented here (see Equations 2 to 4 and the coefficients and prediction intervals in Table 3), corrected values and their estimated accuracy can be calculated. These models

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FIGURE 7 Comparison of VSS speed versus 5th wheel measured speed (left) (diagonal lines represent a 1:1 relationship) and difference between VSS speed and 5th wheel speed as a function of 5th wheel acceleration (right). Top figures show data from PCS events. Bottom figures show data from ABS events with data points grouped by ABS intervention.



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improve the utility of VCH data for accident reconstruction purposes by reducing error and quantifying accuracy.

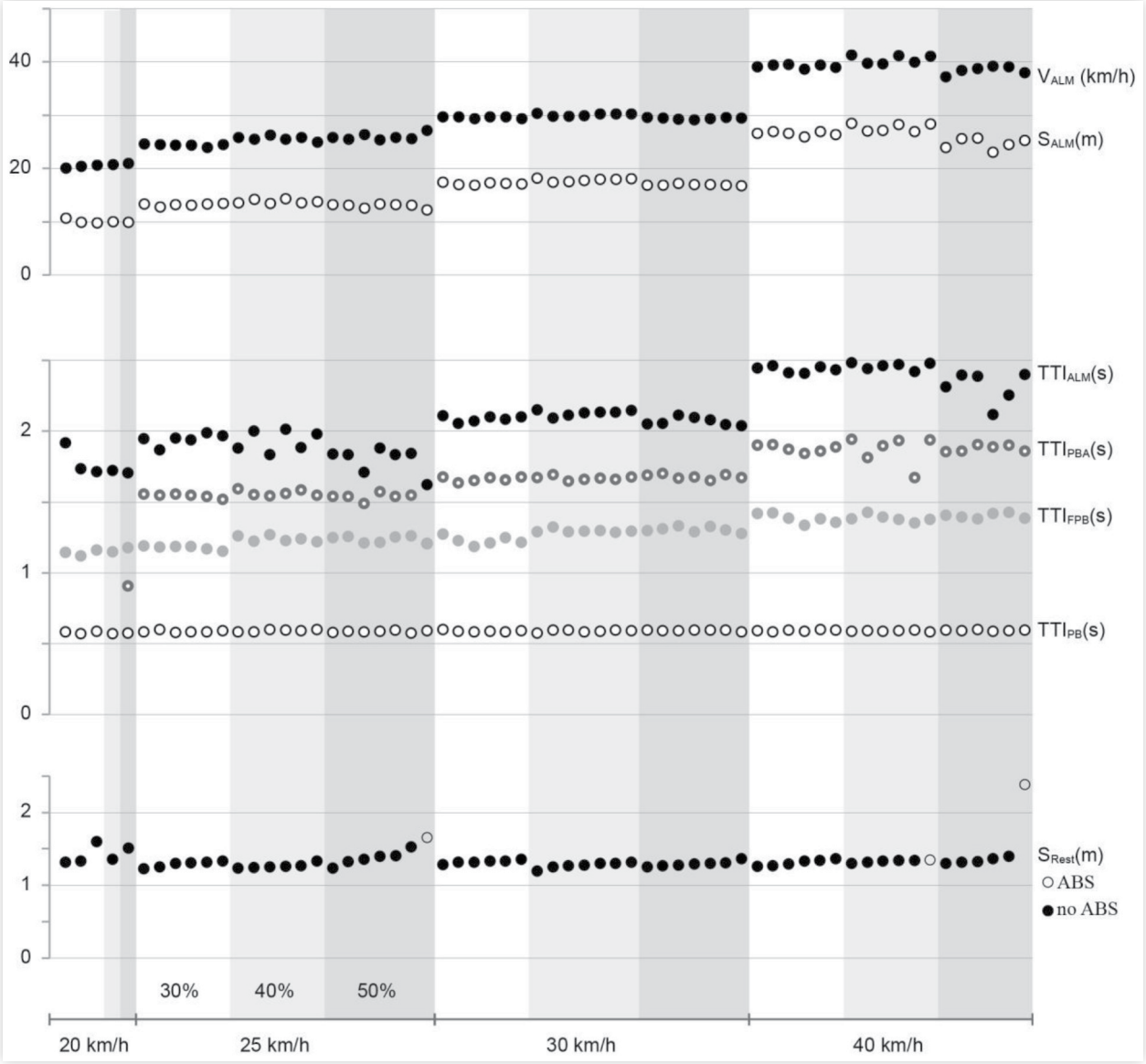
For both PCS and ABS events, the VSS consistently underestimated the reference 5th wheel speed, but was generally within 2 km/h (Figure 7). The errors increased with increased braking level and the largest errors occurred during ABS activation. The VSS underestimated the reference speeds by -0.43 ± 0.58 km/h in PCS records and by -0.29 ± 0.58 km/h (without ABS intervention) in ABS records. For test speeds between 20 and 40 km/h without ABS intervention, this represent an underestimate between 0.7 and 1.4%, which is consistent with the vehicle CAN bus speed accuracy found by Brown and White [9]. During heavy braking, the maximum VSS error was -8.1 km/h for PCS records and -11.2 km/h for ABS records, which is consistent with the findings of Ruth et al [8]. These larger errors may be related to tire slip causing underrotation of the wheels and possible timing differences between the CAN bus data and the VCH recording of that data.

The longitudinal G sensor (A_{SENSOR}) underestimated the vehicle's acceleration (overestimated the vehicle's deceleration) by an average of 0.063 m/s^2, but this error was small compared to the distribution of the data and the size of the prediction interval produced by the linear regression analysis. During hard braking the sensor saturated at -8.9725 m/s^2, and therefore this value should not be considered an accurate indication of the instantaneous acceleration of the car.

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FIGURE 8 Summary of 62 matrix tests that did not result in a target strike. The tests are grouped according to test speed, then sub-grouped according to accelerator pedal position (shaded vertical columns), and then sorted within these subgroups according to distance at rest from the target. Note that there were four 20-km/h tests in which there were no PBA, and one 20-km/h test in which PBA occurred after FPB.



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With respect to the effect of initial vehicle kinematics on PCS performance, we found that test speed and accelerator position affected most of the PCS data we analyzed, although some of these differences were small from a practical perspective (Table 4). Overall, changes in test speed had a larger effect than changes in accelerator position on most of the PCS variables analyzed.

Despite doubling the test speed from 20 km/h to 40 km/h, we observed only a 37% increase in TTI_{ALM}, a 21% increase in $TTI_{FPB,}$ and no difference in the distance to target at rest (Table 4). Because the TTIs did not double along with the test speed, there may be a speed above which there is insufficient time to stop before the target under the friction conditions present during our tests. Under lower friction conditions, the TTIs we measured may be insufficient to allow the vehicle to stop before the target, even at the speeds we tested. Further work is needed to explore the PCS performance at higher speeds and/or under lower friction conditions.

The Insurance Institute for Highway Safety (IIHS) awarded a top score to the 2017 Toyota Corolla's PCS because it stopped short of their target in 20 km/h (12 mph) and 40 km/h (25 mph) tests. The IIHS test protocol [12] specifies a 150 and 200 m initial distance, a slow acceleration to the test speed, and a final constant-speed section of 30 m and 60 m for the 20 km/h and 40 km/h tests, respectively. The results of our long approach tests at 20 km/h are consistent with the good results of the IIHS tests. However, in 13 of our 18 short-approach tests at 20 km/h, the vehicle struck the target. These results suggest that the PCS may not work as well in stop-and-go traffic as it does in the IIHS tests.

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TABLE 4 P-values (top rows) and post-hoc means ± standard deviations (bottom rows) for the effects of test speed and accelerator opening ratio on the dependent variables. For each dependent variable, the post-hoc values are significantly different from one another unless indicated by common superscript letters.

Variables		N	V_{ALM} (m/s)	S_{ALM} (m)	TTI_{ALM} (s)	TTI_{PBA} (s)	TTI_{FPB} (s)	TTI_{PB} (s)	S_{Rest} (m)
Test Speed		62	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	0.0009	0.1457
Accelerator Opening Ratio		62	<0.0001	<0.0001	<0.0001	0.1209	<0.0001	0.8751	0.0312
Post-hoc results									
Test Speed (km/h)	20	5	5.69 ± 0.10	10.00 ± 0.36	1.759 ± 0.089	1.599 ± 0.395[a]	1.151 ± 0.022	0.579 ± 0.007	142 ± 13 [a]
	25	19	7.03 ± 0.23	13.24 ± 0.48	1.886 ± 0.100	1.531 ± 0.082[a,b]	1.219 ± 0.035	0.589 ± 0.007[a]	133 ± 11 [a]
	30	20	8.23 ± 0.10	17.26 ± 0.46	2.096 ± 0.035	1.670 ± 0.017[b]	1.280 ± 0.041	0.591 ± 0.006[a]	130 ± 4 [a]
	40	18	10.92 ± 0.30	26.25 ± 1.47	2.403 ± 0.092	1.875 ± 0.061	1.391 ± 0.027	0.592 ± 0.005[a]	139 ± 25 [a]
Accelerator Opening Ratio (%)	30	21	8.18 ± 1.94	17.65 ± 6.19[a]	2.104 ± 0.240[a]	1.708 ± 0.141[c]	1.246 ± 0.097	0.589 ± 0.007[b]	132 ± 7[b,c]
	40	20	8.71 ± 1.83	19.13 ± 6.12	2.150 ± 0.239[a]	1.699 ± 0.134[c]	1.299 ± 0.070[a]	0.590 ± 0.007[b]	129 ± 4[b]
	50	21	8.44 ± 1.56	17.43 ± 5.10[a]	2.030 ± 0.234	1.637 ± 0.240[c]	1.305 ± 0.076[a]	0.591 ± 0.007[b]	141 ± 25[c]

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TABLE 5 Radar warning test results, showing the mean ± standard deviation of the dependent variables for the far, middle, and near radar warning settings. The * indicates conditions that are significantly different from one another.

Variables	Far	Middle	Near	p-value
V_{ALM} (m/s)	11.338 ± 0.274	11.158 ± 0.440*	11.813 ± 0.130*	0.0056
S_{ALM} (m)	30.278 ± 1.204*	27.577 ± 1.657*	25.518 ± 1.292*	0.0001
TTI_{ALM} (s)	2.670 ± 0.044*	2.470 ± 0.053*	2.161 ± 0.115*	<0.0001
TTI_{PBA} (s)	1.923 ± 0.030	1.907 ± 0.047	1.894 ± 0.086	0.6696
TTI_{FPB} (s)	1.392 ± 0.027	1.388 ± 0.029	1.389 ± 0.014	0.9511
TTI_{PB} (s)	0.588 ± 0.003	0.592 ± 0.006	0.593 ± 0.006	0.3368
S_{Rest} (cm)	134.75 ± 5.71	131.39 ± 5.93	135.98 ± 9.39	0.4347

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The Toyota Corolla's owner's manual states that the pre-collision brake assist operates between approximately 30 and 180 km/h. Nevertheless, we found PBA flags in 58% of the 20 km/h tests and all of the 25 km/h tests. Therefore, PBA can activate below 30 km/h and the presence of a PBA flag cannot be used to infer an initial speed of greater than 30 km/h.

When we combine the 24 short- and long-approach tests at 20 km/h, we observed 13 target strike tests with PBA flags present, 10 non-strike tests when the PBA flag was absent, and only one non-strike test when the PBA flag was present. This latter test was the only test in which the PBA flag occurred after the FPB flag. There appears to be a correlation between the presence of a PBA flag and target strikes; however, we were unable to discern an underlying cause.

With respect to the third goal, we found that the "pre-collision warning timing" setting affected the timing of the ALM flag, which corresponds to driver warnings, but did not affect the other PCS variables we investigated. Adjusting the "pre-collision warning timing" did not change the braking behavior of the PCS, but did change how far from a potential impact the warning buzzer and the instrument panel "BRAKE" message appeared.

In some tests, the radar-measured target distance and target relative velocity demonstrated sudden and short-term changes before returning to values that agreed with the 5th wheel values (e.g. see first 3 data points in Figure 5). These changes occurred simultaneously in both variables for at most 1.5s. At these discrete changes, the PCS "Target Number" variable in the VCH data indicated a number greater than 1, suggesting that the system was actively tracking multiple targets and had temporarily prioritized a different target. These target changes typically occurred in the first few seconds of the VCH data in our tests (most often when the vehicle was still stopped), and we observed only four instances of target prioritization loss while the vehicle was moving.

The tests performed in this study were limited to a single 2017 Toyota Corolla, a narrow range of approach speeds and acceleration levels, and a single target. The results and PCS behavior may not be applicable to other vehicles, different speeds and accelerations, and different targets. The tests were performed using a stationary AEB target and the PCS system may behave differently with a moving target. All of our tests were conducted on the same dry asphalt surface, and the results may change with different road conditions, particularly those with a lower available friction. Despite these limitations, we were able to observe the Toyota PCS stop the vehicle in a relatively consistent manner across a range of speeds, accelerations, and warning settings.

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Conclusions

We quantified the accuracy of vehicle speed, vehicle acceleration, radar target closing speed, and radar target distance read from the VCH of a 2017 Toyota Corolla following AEB events. We derived linear regression equations that predicted reference distance, speed, and acceleration values to within 1.00 m, 0.46 m/s, and 0.42 m/s^2, respectively. These equations improve the utility of VCH data for accident reconstruction purposes by reducing error and quantifying accuracy.

We found the vehicle speed sensor underestimated the reference speed between 0.7% and 1.4% when the ABS was not active. We also found that the vehicle speed sensor underestimated the reference speed by up to 11.2 km/h when the ABS was active. Both findings are consistent with previous research.

The time-to-impact (TTI) when the 4 stages of the PCS occurred (alarm, pre-collision brake assist, former pre-collision brake, and pre-collision brake) varied primarily with test speed, and varied to a lesser extent with accelerator pedal position. The magnitude and variation of the TTIs suggest that the PCS's ability to stop short of a stationary target may be reduced at higher speeds and/or lower friction conditions than we tested.

Adjusting the "pre-collision warning timing" between "near", "middle", or "far" affected only when the warning buzzer and instrument panel "BRAKE" message appeared; it did not affect the braking performance of the PCS.

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Definitions/Abbreviations

AEB - Autonomous Emergency Braking
ALM - PCS Alarm
CAN - Controller Area Network
FPB - Former Pre-collision Brake
PB - Pre-collision Brake
PBA - Pre-collision Brake Assist
PCS - Pre-Collision System
TTI - Time To Impact
VCH - Vehicle Control History
VSS - Vehicle Speed Sensor

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Appendix A: VCH Data List

Labels marked with * denote data available in PCS events.

Time (sec)*
Data ID*
Trigger ID*
Trigger ID (Sub)*
Trigger Time Information (ms)*
Detection Counts of Certain Level of Acceleration (ms)*
Detection Counts of Certain Level of Acceleration*
Number of Overwrite of the Recording Area*
Time and Date*
Key Cycle (trip)*
Clock Type*
Elapsed Time (ms)*
ODO*
Odometer Unit Signal*
HV Recognition Flag_Header*
Diesel Flag_Header*
Writing Completion Flag*
Successive Trigger Time Information (ms)
Vehicle Speed (km/h)*
Accelerator Opening Ratio (%)*
Engine RPM Data (rpm)*
Cruise Control*
Shift Position Signal*
Brake SW*
Brake Oil Pressure (MPa)*
Yaw Rate Sensor Signal (deg/s)*
Longitudinal G Sensor Signal (m/s2)*
Lateral G Sensor Signal (m/s2)*
Throttle Opening Ratio (%)*
Command Value of Fuel Injection Amount (mm3/st)*
Steering Signal (deg)*
Shift Gear
PWR Mode
ECO Mode
Integrated Sport Mode Indicator
SNOW Mode
Valid/Invalid of Driving Mode
VSC OFF Lamp
TRC OFF Lamp
Valid/Invalid of TRC/VSC OFF Lamp
EV Mode
Valid/Invalid of EV Mode
Integrated Sport Mode Signal
Engine Load Factor (%)
READY Signal
Gear Position Signal
Turbine RPM/Input Shaft RPM (rpm)
Under VSC Control
Under ABS Control
Under VDM Control
Under TRC Operation Flag
FR Wheel Speed (km/h)
FL Wheel Speed (km/h)
RR Wheel Speed (km/h)
RL Wheel Speed (km/h)
Trigger Judgment Information
Cruise System
Driving Force Demand Judgment Flag
Driving Force Demand Judgment Flag
Cruise Control Master Switch Flag
DSS Judgment Flag
DSS Judgment Flag
Cruise Control Res/Accel SW
Cruise Control Res/Accel SW
Cruise Control Set/Coast SW
Cruise Control Set/Coast SW
Cruise Control Cancel SW
Cruise Control Cancel SW
Cruise Control Brake SW
Cruise Control Brake SW
Cruise Control D Signal
Cruise Control D Signal
Cruise Control Main SW
Cruise Control Main SW
Cruise Control Set Speed (km/h)
Cruise Control Set Speed (km/h)
Vehicle Speed (during Cruise Control) (km/h)
Vehicle Speed (during Cruise Control) (km/h)
Cruise Control Request Value (%)
Cruise Control Request Value (%)
Cruise Control Request Value (N)
Cruise Control Request Value (N)
ACC Target Distance (m)
ACC Target Distance (m)
ACC Target Condition
ACC Target Condition
ACC Relative Speed (km/h)
ACC Relative Speed (km/h)
PCS Reception Data Invalid Flag*
ALM Request Flag*
ABK Request Flag*
PBA Request Flag*
FPB Request Flag*
PB Request Flag*
Extension Request Flag*
PSB Request Flag*
PCS READY Flag*
PCS Display (Temporarily not available)*
PCS Display (Radar Dirt)*
PCS Display (System Fault)*
PCS Display (PSB Communication)*
Target Distance (m)*
Target Relative Velocity (m/s)*
Target Lateral Position (m)*
Target Number*
Detection of Certain Level of Acceleration*
LKA DDR Trigger Signal
Invalid of LKA DDR Data
Steering Wheel Release Condition
Steering Operation of LKA

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Steering Operation of LDW/LDA
LKA Control Condition (EPS Control)
LKA Control Condition
Turn Signal Operation
EPS Torque Sensor Value (Nm)
HV Recognition Flag*
Diesel Flag*
Sensor Type Identification (Security)*
Sensor Type Identification (G Sensor Layout)*
Sensor Type Identification (G Sensor)*
Sensor Type Identification (Yaw Rate Sensor)*
Yaw G Sensor Installation Information

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